ACME METALS

Digital Material Passport

ID **DMP-METAL-005** - *Version* **1.0.0**
Issue Date **2025-05-17** - *Certificate Type* **EN 10204 3.1**

Manufacturer	Customer	Business Transaction
ACME Metal Works GmbH Industrial Park 123 52066Aachen DE quality@acme-metal.example.com	**European Research Institute** Science Boulevard 42 75015Paris FR materials@eri.example.org	*Order ID* **PO-23456** - Pos. 1 - 2025-05-01 - Quantity **1000 kg** *Delivery ID* **DN-65432** - Pos. 1 - 2025-05-16 - Quantity **1000 kg**

Product

Product Name	Titanium Alloy Ti-6Al-4V ELI	*Number (UNS)*	R56401
Batch ID	T-65432-01	*Shape*	RoundBar - D 30 mm - Length 3000 mm
Surface Condition	Machined	*Product Norm*	ASTM F136 (2022)
Weight	1000 kg		
Production Date	2025-05-15		
Country of Origin	DE		

Chemical Analysis

Heat Number **T-65432** - *Melting Process* **VAR** - *Casting Method* **VacuumCasting** - *Casting Date* **2025-05-14** - *Sample Location* **Product**

Symbol	*Unit*	*Min*	*Max*	*Actual*
Ti	%			89.32
Al	%	5.5	6.5	6.02 ± 0.05
V	%	3.5	4.5	3.95 ± 0.03
Fe	%		0.25	0.18

Symbol	*Unit*	*Min*	*Max*	*Actual*
O	%		0.13	0.11
C	%		0.08	0.026
N	%		0.05	0.012
H	ppm		120	35

Mechanical Properties

Property	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*
Fatigue Test Room temperature, R = 0.1				ASTM E466	✓

Cycles (N)	**10000**	**50000**	**100000**	**500000**	**1000000**	**10000000**
Value	650	635	610	590	570	550

Property	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*
Notch Sensitivity	0.85 - 0.92			Internal Method TS-5432	✓

Physical Properties

Property	*Actual*	*Target/Min*	*Maximum*	*Method*	*Status*
Density	4.43 ± 0.01 g/cm³	4.43 g/cm³		ASTM B311	✓

Supplementary Tests

Test	*Result / limits*	*Method*	*Status*
Microstructure Examination	Equiaxed alpha with intergranular beta	ASTM E407	✓
Ultrasonic Inspection	Yes - No indications greater than reference standard	ASTM E2375	✓
Surface Quality Assessment	Class 1 - Medical Grade	Visual Inspection per ASTM F136	✓
Alpha Case Depth	5 µm - max 25 µm	Microhardness Traverse	✓
Grain Size Distribution	8 - 10 ASTM No. - target 7 - 12 ASTM No.	ASTM E112	✓

Hardness Profile (ASTM E384), values in HV

Distance from surface (mm)	**0.1**	**0.5**	**1**	**2**	**3**	**5**	**10**	**15**
Value [HV]	345	350	352	350	349	348	351	347

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with ASTM F136 and meets all requirements for surgical implant applications.

Validated by **Dr. Markus Weber**, Head of Metallurgy, Research & Quality - 2025-05-17